

Lesson Exemplar in General Science

Quarter 2

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Lesson Exemplar for General Science
Quarter 2: Unit 3

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LESSON EXEMPLAR			
Learning Area	General Science	Grade Level	11
Semester	1 st	Quarter	2

I. OBJECTIVES (*Identifying the Goals*)

Content Standard	<i>The learners learn that chemical reactions are part of human activities and present in the environment;</i>
Performance Standard	<i>By the end of the quarter, learners will apply research skills to find out how scientists have made an impact on our lives. They recognize the significant role of different agencies in the Philippines, such as DOST and DTI, in scientific innovation and development. They show critical thinking in explaining the different substances present in household and personal care products including their benefits and potential risks of using and disposing of them. They demonstrate appropriate and safe handling and use of chemicals. They explain that chemical reactions take place in human activities and in the environment, and that there are several chemical solutions that pose a threat to the community.</i>
Learning Competencies	Prior Knowledge and skills Learned in the Previous Grade Level: <i>The students in Grade 10 learned about several simple observations that indicate if a chemical reaction has taken place. They described the indicators for a chemical reaction as color change, the formation of a precipitate, the release of gas, and or odor, or a change in temperature with acids, bases, and salts. They also described important types of chemical reactions (combination, decomposition, single replacement, double replacement). Many types of chemical reactions are important in our daily lives and in the biotic and abiotic parts of the environment.</i> Learning Competencies: <i>The learners shall be able to:</i> <i>explain some of the common types of chemical reactions and provide examples for the following: decomposition, acid on carbonates, acids on metals, and combustion;</i> <i>identify the type of chemical reactions that occur in our daily activities including in baking, cleaning, and burning;</i>

II. REFERENCES and MATERIALS

(Selecting Resources and Material)

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Sources of Activities:

- Helmenstine, Anne. "Chemical Equation for Baking Soda and Vinegar Reaction." Science Notes and Projects, July 4, 2020. <https://sciencenotes.org/chemical-equation-for-baking-soda-and-vinegar-reaction/>.

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<https://www.zoomdrain.com/blog/2022/june/will-vinegar-and-baking-soda-unclog-a-drain/>

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Worksheets/learning activity sheet (LAS)
 Presentation slides
 Materials for the *magic tricks*

(These shall be accomplished per topic)

III. CONTENT <i>(Sequencing Content)</i>	<i>This section outlines the lesson or topic that learners need to learn to independently attain the Content Standard and achieve the Performance Standard</i> Chemical reactions in everyday lives 1.Types of chemical reactions 2.Evidence and chemical equations of chemical reactions 3.Examples of chemical reactions that occur in daily activities, including baking, cleaning, and burning 4.Sample experiments that demonstrate the types of chemical reactions	
IV. OBJECTIVES <i>(Setting Clear Objectives and Analyzing the Tasks)</i>	Lesson Objectives: <i>Learners shall be able to:</i> 1. <i>Classify different types of chemical reactions, including synthesis, decomposition, acid-carbonate, acid-metal, and combustion, by identifying their general reaction forms, reactants, and products.</i> 2. <i>Conduct simple experiment involving products, to recognize the chemical reactions involved in practical scenarios.</i> 3. <i>Provide real-life examples of various types of chemical reactions and explain their relevance in everyday life.</i>	
IV. PROCEDURES <i>(Selecting Strategies, Making Meaningful Content, Delivering Lesson and Assessing Learning)</i> This section focuses on selecting learner-centered, evidence-based instructional approaches such as problem-based learning, collaborative tasks, interdisciplinary integration, and technology-enhanced instruction. These strategies are intended to foster active engagement, critical thinking, and adaptability across diverse learning pathways. The chosen approaches and methodologies will be reflected through varied and relevant activities and assessments that emphasize real-world relevance and application, thereby enhancing learner engagement and comprehension. (Each part shall have 2-3 varied activities)		ANNOTATION <i>*Instruction to the teacher on how to facilitate the activities.</i> <i>*In the Annotation, explicitly explain how the IDF is applied in each part of the lesson</i>
A. Activating Prior Knowledge	1. Activating Prior Knowledge 1. Short Review of the concept of chemical reaction and the indicators that a chemical reaction has occurred Option 1: Gallery Walk Divide the class into 5 groups. Display photos that demonstrate chemical reactions. Have the students do a gallery walk.	Review the concept of chemical reaction and indicators that a chemical reaction has occurred. Option 1. Note to the teacher: Put up the images or pictures in strategic corners in the room. images or pictures of different signs of



Figure 1. Burning matchstick (Helmenstine, 2020)

The teacher should provide the students a mini-worksheet with the guide questions.

Guide Questions:

1. What observable change can you identify in the pictures?
2. What is common among all these pictures ?
3. How do we know that a chemical reaction or a chemical change took place?
4. Are all reactions the same? Why or why not?

chemical reaction or chemical change (evolution of gas, light, change in color/odor, etc. A sample picture is found in figure 1. Ask the guide questions for processing.

Note: When the modality shifts to blended, this can be done by displaying the picture in the class point with word cloud feature when in synchronous class or convert this into mini-worksheet for asynchronous or modular.

IDF RELEVANCE

The Gallery walk point to the reflective and relevant part of the framework. The teacher helps the students to think about what they already know before the lesson. As each group of students look at the picture in each part of the gallery, they ponder on the question given and reflect on how it relates to what they have learned before.

It is reflective as the teacher thinks about what students already know and how to build on it. Overall, the experiment helps spark curiosity, connects with past learning, and prepares students for deeper understanding.

Note to the teacher: The teacher must link the last question to types of chemical reaction. Recall concepts in Grade 10

	Option 2: Simple Demonstration /Chem magic Demonstrate to the students simple experiments that manifest signs of chemical reaction or chemical change (evolution of gas, light, change in color/odor, etc.) Guide Questions: 1. What observable change can you identify in the simple demonstration? 2. How do we know that a chemical change or a chemical reaction has occurred? 3. Do you know of other chem tricks similar to this? How similar and different is/are the chem tricks with this?	Option 2. Note to the teacher: Perform the experiments in front of the class. The simple experiments should appeal to the students as if the teacher is performing magic tricks. Here are some of the links where tricks can be adopted: Bob Science. 2015. "Rapid Color Changing Chemistry! - ScienceBob.com." ScienceBob.com. February 21, 2015. https://sciencebob.com/rapid-color-changing-chemistry/. Helmenstine, Anne. 2020. "Gatorade and Drain Cleaner Color Change Chem Demo." Science Notes and Projects. February 25, 2020. https://sciencenotes.org/gatorade-and-drain-cleaner-color-change-chem-demo/. Helmenstine, Anne. 2013. "Make Homemade Colored Soap Bubbles." Science Notes and Projects. May 30, 2013. https://sciencenotes.org/colored-soap-bubbles/. IDF RELEVANCE The demonstration also acts as a formative assessment, allowing the teacher to check students' understanding based on their predictions and observations. Through this approach, the Chem Magic demonstration becomes more than just entertaining—it activates prior knowledge and prepares students for deeper learning.
	2. Establishing the Purpose of the Lesson Building upon the review, the teacher will inform learners that lessons will focus on different types of chemical reactions, real-life examples, and their relevance in everyday life	From the review, the teacher will ask probing questions about chemical reactions. The key concept is that there are different types of chemical reaction and

	Essential Questions: 1. How do different types of chemical reactions, such as decomposition and combustion, help us understand changes around us? 2. In what ways do chemical reactions shape our everyday lives/experiences? Option 1: STEAM Vocabulary Introduce the STEAM vocabulary. Ask the students to write 3 words or terms related to the lesson/topic and define/describe or give examples related to the term. Materials Needed: • meta cards or small pieces of paper • Markers or pens Instruction: From the given purpose and goal of the lesson, give at least 3 terms and define/describe or give examples related to the term. Write your answers in a meta-card. Post them in the appropriate column. <table><tr><th>Terms</th><th>Meaning/Description/Example</th></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table> Option: The teacher may make use of class point to solicit the terms along with their meanings, descriptions, and examples. This option can be made for synchronous modality, where the teacher can meet the class via Google Meet, Zoom or Microsoft Teams. Prepare two (2) slides for this. Ask the students to log in	Terms	Meaning/Description/Example							they part of everyday life and play essential roles in daily activities like cooking, cleaning, and other routine activities. Possible key vocabulary words • Reactant • Product • Synthesis/combination • Decomposition • acid-carbonate, • acid-metal • Combustion • gas release • heat • color change • Change in odor • Formation of precipitate Note: When the modality shifts to blended, make use of class point to solicit the terms, Meaning/Description/Example. This option can be made for synchronous modality, where the teacher can meet the class via Google meet, Zoom or Microsoft Teams. Follow the same preparation as stated in the option. IDF RELEVANCE The STEAM Vocabulary activity applies this IDF framework by encouraging students to think deeply and connect their prior knowledge to the new lesson. It is relevant when students describe or give examples of the terms based on their own
Terms	Meaning/Description/Example									

	the Class Point. For the first slide, the teacher will insert a word cloud to solicit terms from the students. The students will type their answers, which will then be displayed on the PowerPoint slide. The second slide will use the short-answer format, where students will provide the meaning, description, or example of the terms. For discussion purposes: Once the table is complete, review all the vocabulary words together. Ask the students to discuss their meanings, share their ideas, and explain the importance of each word. Encourage active participation and provide clarification for any misconceptions, offering additional explanations or examples if necessary. Option 2: Anticipation Guide Introduce the anticipation guide to activate students' prior knowledge and build curiosity about the different types of chemical reactions, their real-life examples, and their relevance in everyday life. Students should indicate whether they agree or disagree with the statements both before and after the instruction.	understanding or real-life experiences. It is responsive when the teacher checks students' answers and gives feedback or clarification based on what they wrote, adjusting instruction to meet their needs. The anticipation guide is enclosed to this lesson. If this is used in the lesson, make sure that the anticipation guide is also used in evaluating learning
B. Instituting New Knowledge	Day 1 1. Presenting Examples Option 1. The Reaction Stations Divide the class into 5 groups. Each group will be assigned a station that represent type/s of reaction. Each group will perform the experiment in each station in 7-10 minutes. Station 1: (Synthesis) Steel wool-soaked in warm saltwater is exposed to air Station 2: (Decomposition) When baking soda is heated Station 3: (acid-carbonate) React vinegar (acetic acid) with baking soda (sodium bicarbonate) to observe an acid-carbonate reaction.	For Presenting Examples, there are 2 option presented. The teacher may choose from the option for presenting Examples. Note to the Teacher: Materials and activity sheet will be provided for each station. Enclosed in this lesson is the activity for Reaction Station IDF Relevance. The Instructional Development Framework (IDF) is applied by making the activity reflective, relevant, and responsive. It is reflective as the students recall prior knowledge about chemical reactions, discuss the results and compare them with their expectations, which in turn encourage reflection on misconceptions or prior learning. It is relevant

	Station 4: (acid-metal) When an acid (vinegar/calamansi juice) reacts with aluminum foil. Station 5: (combustion) Burning a small alcohol-soaked cotton ball under controlled conditions After the experiment. The student will complete the table. <table><tr><th>Station</th><th>Materials Used (reactants)</th><th>Observation</th><th>Product</th></tr><tr><td>1</td><td></td><td></td><td></td></tr><tr><td>2</td><td></td><td></td><td></td></tr><tr><td>3</td><td></td><td></td><td></td></tr><tr><td>4</td><td></td><td></td><td></td></tr><tr><td>5</td><td></td><td></td><td></td></tr></table> Guide Questions: 1. What are your observation in the different stations? 2. How do these changes help you identify the type of chemical reaction?	Station	Materials Used (reactants)	Observation	Product	1				2				3				4				5				because the experiments use familiar, safe materials that students may have seen at home or in daily life. Also, the instructions and tasks are aligned with curriculum standards and learning goals related to reaction types Note to the teacher: The teacher may use the attached rubric or may involve the students on how they will be graded. The rubrics attached was modified from an AI generated rubrics
Station	Materials Used (reactants)	Observation	Product																							
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	Option 2: Predict-Observe-Explain Divide the class into 5 groups. Each group will be given materials they need to design an experiment that represent type/s of reaction. Each group will be perform POE on at least 1 safe and home experiments that demonstrate different types of chemical reactions: <i>synthesis, decomposition, acid-carbonate, acid-metal, and combustion.</i> <table><tr><th>Experiment</th><th>Materials Used (provide description/characteristics)</th><th>Predict the Product (Note the color, light, and type of</th><th>Observation</th><th>Explanation</th></tr><tr><td></td><td></td><td></td><td></td><td></td></tr></table>	Experiment	Materials Used (provide description/characteristics)	Predict the Product (Note the color, light, and type of	Observation	Explanation						Note to the Teacher: Materials to be provided should be varied to cover the different types of chemical reactions: <i>synthesis, decomposition, acid-carbonate, acid-metal, and combustion the experiments in the Reaction station may be used in this POE (Predict-Observe-Explain) activity but restructured to suit the POE</i> IDF Relevance IDF (Instructional Development Framework) is effectively applied in the POE (Predict-Observe-Explain) strategy. Through POE, students use their past knowledge to make predictions about what they believe will occur in an experiment or														
Experiment	Materials Used (provide description/characteristics)	Predict the Product (Note the color, light, and type of	Observation	Explanation																						

	<table><tr><td></td><td></td><td>material formed.)</td><td></td><td></td></tr><tr><td>1</td><td></td><td></td><td></td><td></td></tr><tr><td>2</td><td></td><td></td><td></td><td></td></tr><tr><td>3</td><td></td><td></td><td></td><td></td></tr><tr><td>4</td><td></td><td></td><td></td><td></td></tr><tr><td>5</td><td></td><td></td><td></td><td></td></tr></table> Guide questions: 1. How did the actual results compare to your predictions? What did you learn from this? 2. Why do you think the reaction happened the way it did? 3. Can you explain why your prediction was correct or incorrect? Optional: To prepare the students for the discussion of the different types of reactions, the teacher may give a reading assignment about the type of chemical reaction focusing on the key concepts.			material formed.)			1					2					3					4					5					scenario. They compare their predictions with the actual results after seeing the outcome, which promotes reflective process. Learning becomes more interesting when students can apply what they have learned to real-world or practical situations during the explanation phase. By strengthening the link between existing knowledge and new understanding, POE within the IDF framework guarantees that learning is deliberate, goal-oriented, and student-centered. Note to the teacher: The teacher may use the attached rubric or may involve the students on how they will be graded. The rubrics attached was modified from an AI generated rubrics
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	Day 2 1. Discussing New Concept Present the outputs of the activity (either option 1 or option 2). Discuss answer to questions on observed changes (color, temperature, gas released, and others) The teacher will identify the type of reaction or name of the reaction. From the result of the experiment, verify with the students the identified type of chemical reaction through a graphic organizer. Use the questions under each column to help the students identify the reaction and provide a real-life example.	Note to the teacher: the students are presumed to have prior knowledge in writing and balancing equation through Science 10, 4th quarter lesson. IDF Relevance The Instructional Development Framework (IDF) is utilized in this activity through the utilization of a graphic organizer in promoting reflective, relevant, and responsive instruction. It enables students to think critically, organize their knowledge																														

	Reactants	Products	Observable Clues	Type of Reaction	Balanced Chemical equation	Real-Life Application
	What substances were combined or used before the reaction?	What new substances were formed after the reaction?	What signs did you observe? (e.g., bubbles, temperature change, color change)	What type of reaction was observed?	What evidence supports this as a decomposition reaction? Write the balanced equation and identify the products.	Where can this type of reaction be found in daily life?
	Vinegar and baking soda	Carbon dioxide (CO ₂), water (H ₂ O), and sodium acetate	Bubbling/fizzing	Acid-Carbonate Reaction	$\text{NaHCO}_3 + \text{CH}_3\text{COOH} \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{NaC}_2\text{H}_3\text{O}_2$	cleaning agent; unclogging drains
Processing questions: - What do you expect to see in the reactions? - What do you think will prompt you to classify the experiment under each type of reaction? - What evidence supports the reaction? Write the balanced equation and identify the products. - How does understanding the general reaction form help you predict the products of similar reactions in real-life scenarios?						
of chemical reactions, and relate them to practical uses while enabling the teacher to monitor learning and give instantaneous feedback. Note to the teacher: In this part, the teacher will facilitate the discussion after the presentation of the outputs. The teacher may refer to the following outline: 1. Explain the goal: identify the type of chemical reaction based on observations and evidence. 2. Introduce the graphic organizer with columns: Reactants, Products, Observable Clues, Type of Reaction, Balanced Chemical Equation, Real-Life Application. 3. Starting with the first 3 columns, ask: - What substances were combined or used before the reaction? - What new substances were formed after the reaction? - What signs did you observe? (e.g., bubbles, temperature change, color change) 4. For the 4th column, ask: “<i>What type of reaction was observed?</i>” and guide students to consider characteristics of the reaction. 5. Lead the discussion for the Types of Reaction and give examples for each major reaction type. Some references include: Byjus. “Types Of Chemical Reactions.” BYJUS. Byju’s, January 4, 2016. https://byjus.com/chemistry/types-of-chemical-reactions/.						

	4. Which reactions are most common in our daily life? 5. Which of these could have environmental impacts? 6. Can you think of any safety concerns related to these reactions?	Pinigis, Alex. "Types of Chemical Reactions." ChemTalk, February 6, 2021. https://chemistrytalk.org/types-of-chemical-reactions/. Soult, Amy. "5.3: Types of Chemical Reactions." Chemistry LibreTexts, May 3, 2019. https://chem.libretexts.org/Courses/Valley_City_State_University/Chem_121/Chapter_5%3A_Introduction_to_Redox_Chemistry/5.3%3A_Types_of_Chemical_Reactions 6. Relate experiment clues/observation from the experiment to the reaction type. Ask: - What do you think will prompt you to classify the experiment under each type of reaction? - What evidence supports the reaction? Write the balanced equation and identify the products. 7. Guide students in writing and balancing the chemical equation from the experiment. 8. Connect discussion to Real-Life. Ask - Where can this type of reaction be found in daily life? Provide or elicit examples such as cleaning agents, digestion, rusting, combustion engines. 9. Extend discussion. Ask: - How does understanding the general reaction form help you predict the products of similar reactions in real-life scenarios?
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		- Which reactions are most common in the our daily life? - Which of these could have environmental impacts? - Can you think of any safety concerns related to these reactions?
	3. Developing Mastery Option 1: Matching Activity Provide a mix of reaction scenarios and ask the students to match each item with the correct reaction type. Have them place the items in the appropriate column of the chemical reaction chart and justify their reasoning. Bleach is used to clean a stain When hydrogen peroxide is applied to a wound When rust forms on iron tools left in the rain Antacids for acid reflux A banana turns brown when left When silver jewelry is left out, it tarnishes barbecue grill When eggshells (mostly calcium carbonate) are soaked in vinegar, they dissolve Spoiling of food Vinegar (acetic acid) reacts with basic substances like soap scum or hard water deposits in the bathroom or kitchen, helping dissolve them. baking cakes or cookies, the released carbon dioxide (CO₂) gas helps the batter rise and become fluffy.	IDF Relevance Matching and explaining activities relate to the Instructional Development Framework (IDF) through: (1) students explaining their matched1) students process and reflect on why they made those connections by explaining their matches; (2) the activity gains significance when the items being matched (such as chemical reactions, real-life applications, or scientific terms) are related to the students' everyday experiences or current lessons; and (3) the teacher can modify instruction based on the students' performance, provide clarification, or offer targeted support, making learning more personalized. It is extended and made meaningful in option 2 by a game-based activity. Note to the Teacher: In Chem Feud, the teacher may made use of the items in option 1.

	Synthesis	Decomposition	acid-carbonate	Acid-metal	Combustion
Option 2: Chem Feud Divide the class into teams. The teacher will act as a host. The host will draw a reaction scenario card from a shuffled deck. (<i>Use the reaction scenario in Option 1</i>) A representative of the group will play one at a time. The representative of the group to be the first to hit the buzzer shall be given the chance to answer. He/She must: Name the reaction type (Synthesis, Decomposition, Acid-Carbonate, Acid-Metal, or Combustion) Give a quick justification (10–20 seconds) If correct, the team earns 1 point for the correct type and 1 bonus point for a clear justification. If a team answers incorrectly, other teams may "steal" the question for a bonus point. The group with the highest points earned wins the chem feud.					

C. Demonstrating Knowledge and Skills	Day 3 1. Finding Practical Application Option 1: (Individual) Reaction Detective: “Where do you see these reactions in daily life?” Students will analyze a household or personal care product to understand how its chemical ingredients work, processes or activities at home that demonstrate the type of reactions they are involved in, and their environmental implications. Instructions: 1. Selection: Identify a common household/personal care product, such as: Shampoo, Toothpaste, Bleach, Vinegar, Air freshener, Baking soda, Muriatic acid Identify processes or activities at home (where the household/personal care product was used) that demonstrate the type of reactions 2. Task A. Read about the product to describe the process/activities at home where the product was used. B. Answer the following: 1. What happens when we use this product? 2. What type of chemical reaction is it involved in the use of the product? (e.g., neutralization, decomposition)? / 3. What is the importance of the activity/process that uses the product in our daily life?	Note to the Teacher: The rubrics for the presentation should be provided. The teacher may present previously crafted rubrics or may involve the students on how they will be graded. IDF Relevance: This activity supports IDF by encouraging thoughtful analysis, real-world connections, and adaptive teaching that meets students’ learning needs. The students reflect on common household processes (like cleaning or cooking) and connect them to specific types of chemical reactions. By exploring environmental implications, students see the importance of chemistry in responsible decision-making and daily living. Note to the teacher: The teacher may use the attached rubric or may involve the students on how they will be graded. The rubrics attached was modified from an AI generated rubrics
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	3. Valuing Explain benefits and potential risks of using them (product) or doing them (process/activities) Output: a simple presentation with visuals/slides that contain the following: - Selected product - Household Process/Activity (with description) - Answers to questions Present the output to the class and discuss the guide questions. Option 2: The Art of Chemical Reaction (Chemical Reaction Painting; Individual) This activity will involve art creation from a chemical reaction. Students will observe how acid-base reactions can create visual effects, and use this reaction to make creative artworks. Divide the class into groups and give each group a worksheet for the art activity. Allow 20–30 minutes for the creation of their artwork. Post the outputs for a gallery walk. Gallery Walk (10 minutes): Have the group post the output. Distribute to the groups sets of star stickers. Assign one member to introduce the output briefly to the viewers. The viewers may opt to give the output star stickers. Processing question: - How did the reaction affect your design? - How does understanding chemical reactions enhance creativity?	Note to the Teacher: The teacher may go to this source link to prepare for this activity: Kelly. 2020. "Fizzy Art - Baking Soda Paint." Typically Simple. April 20, 2020. https://typicallysimple.com/fizzy-art-baking-soda-paint/. The teacher may present previously crafted rubrics or may involve the students on how they will be graded. It is a must that the rubric contain the criteria for grading the visual/art. The rubrics attached was modified from an AI generated rubric. Extension Activity: The class may hold one-day exhibit for other class to view the art work. For this, the display should include short explanation about the artwork. The students who will view the art work may also be allowed to judge using a rubric provided. IDF Relevance:
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	2. Making Generalization <i>Option 1: Q and A</i> Ask the students to write one question to draw or summarize what has been learned. Divide the class into 4 groups. Ask the students to share the questions in the group. They may also try to answer the questions in the group. Let each group determine one question they would like to pose. When all the groups are ready, ask each group to pose their question one group after the other. Ask some volunteers to answer the questions. Option 2: Ask essential questions: 1. How can you identify the type of chemical reaction occurring? 2. In what ways do chemical reactions play a role in everyday activities like cooking, cleaning, or using personal care products?	The Art of Chemical Reaction activity applies the Instructional Development Framework (IDF) by (1) Students comprehend how their artistic decisions were influenced by chemical interactions. (2) The processing questions foster metacognition by encouraging more in-depth consideration of the relationship between creativity and science. (3) Students realize the usefulness of chemistry in unexpected, everyday forms when scientific learning is demonstrated through a creative medium. IDF Relevance: The Instructional Design Format supports generalization by providing structured opportunities for learners to reflect on and consolidate their understanding. Q and A, facilitates active student engagement by encouraging them to formulate and discuss their own questions, promoting critical thinking, and collaborative meaning-making. Essential questions, guides students to connect the lesson—such as identifying and applying chemical reactions—to real-world contexts and personal experiences. Learning becomes student-centered, inquiry-driven, and anchored in real-life relevance.
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	3. How does understanding chemical reactions help us in solving everyday problems or making better choices?" Reflective question: <i>How has this week's learning changed your perspective on observing chemical reactions and changes in everyday materials and processes around us?</i> 3. Evaluating Learning Option 1: Anticipation Guide Use the same anticipation guide. Ask the students to answer the 'after' part, and verify with them what changes occurred in their responses. Option 2. Exit Ticket (Quick Check) Answer the following before leaving class: 1. Give and differentiate the different types of chemical reaction. 2. What gas is released when vinegar reacts with baking soda? 3. What is a sign that combustion is occurring? 4. Which reaction involves a metal reacting with acid? 4. Additional Activity/ies (Take home) Individual/Group Activity: ChemVlog Students will analyze a household or personal care product to understand how its chemical ingredients work or process/activities at home that demonstrate the type of reactions they are involved in, and their environmental implications. Instructions: 1. Selection:	IDF Relevance: Clearly defining students' learning requirements assists teachers in developing evaluations that correspond to the objectives. Tools like anticipation guides at the beginning of the lesson serve to assess students' pre-existing knowledge and the post instruction, whereas exit tickets at the conclusion offer brief, reflective glimpses of students' comprehension. These are built into a organized design and provide continuous feedback. IDF Relevance: Creating the vlog requires students to explain their understanding clearly, helping them process and internalize what they've learned. The task connects science to real-life products and daily activities, making learning meaningful and applicable. This becomes reflective and relevant.
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	Observe three activities from your home (e.g., cooking, cleaning, or heating). Describe what happens and identify the type of chemical reaction involved. 2. Video Creation Create a 3–5 minute vlog that includes: • Activity observed • Demonstration or reenactment of the reaction (if safe and possible; you may include members of the family) • Explanation of the type of chemical reaction observed • Description of the visible signs that helped you identify the type of reaction If done as a group, provide a report of the roles and contributions of the members of the group together with the video. Mention the assigned role such as researcher, videographer, presenter, scriptwriter. Show how the group worked together to plan, film, and edit the video. Tips: Creative visuals, props, or analogies to aid understanding Safety measure: Handle household chemicals like vinegar, baking soda and other carefully. Keep sharp objects like nails, knives and others away from children. Perform reaction that might produce spills over a sink or outside.	
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V. ASSESSMENT

- 5-items multiple choices

For question numbers 1-2, kindly refer to the figure.



Figure 2. Baking soda and vinegar (Chemicalbook.com, 2024)

1. A spoonful of baking soda was placed into a glass with vinegar. Which type of chemical reaction occurs when vinegar and baking soda are combined?

- A. Combustion
- B. Acid–Carbonate Reaction
- C. Single Replacement
- D. Decomposition

2. Which of the following balanced chemical equations represents the reaction between vinegar and baking soda?

- A. $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$
- B. $\text{CH}_3\text{COOH} + \text{NaHCO}_3 \rightarrow \text{CH}_3\text{COONa} + \text{CO}_2 + \text{H}_2\text{O}$
- C. $\text{Na} + \text{Cl}_2 \rightarrow \text{NaCl}$
- D. $\text{CH}_4 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$

IDF Relevance:

The quiz emphasize the connection of the content standard- the role of chemical reactions in daily life- by focusing on practical examples (e.g., vinegar and baking soda). The part of investigating the chemical reaction fosters critical thinking and application support by the items in the test -chemical products, equations, and real-life uses.

By directing learning objectives and assessments that correspond to various degrees of cognitive complexity, the SOLO Framework enhances Instructional Design. Teachers can plan instruction to ensure that tasks encourage deeper understanding and direct students from shallow recall to more complex levels of thinking and application. The incorporation of SOLO into Instructional Design allows teachers to develop step-by-step learning activities and assessment for progressive cognitive development, enhancing the effectiveness and clarity of educational interventions.

Note to the Teacher: Under Investigating Chemical Reactions, there are 3 given practical scenario, the teacher may choose 1 or 2 items for the students to accomplish for 5 or 10 points.

Correct Answers:

1. B. Acid–Carbonate Reaction

3. At a campsite, a hiker lights a dry piece of wood on fire. As the fire burns, the wood turns black, releases heat and light, and produces smoke. (Figure 3) Which observation best indicates that a chemical reaction has taken place?

- A. The wood remains the same color and temperature
- B. The fire gives off heat, light, and smoke
- C. The wood cools down over time
- D. The mass of the fire increases

4. You are asked to help a family that has a clogged kitchen sink. They have vinegar and baking soda at home (Figure 4). Why is the reaction between vinegar and baking soda considered useful in such real-life situations?

- A. It produces electricity
- B. It generates light
- C. It can be used as a cleaning agent or to unclog drains
- D. It creates strong adhesives



Figure 3. wood burning in the camp (Evans. 2023)



Figure 4. Vinegar and baking soda to unclog kitchen sink (Zoom Drain Franchise. 2022)

Why? - The reaction between baking soda (sodium bicarbonate) and vinegar (acetic acid) is an acid-carbonate reaction. This is because vinegar, an acid, reacts with baking soda, a carbonate compound, producing carbon dioxide gas, water, and a salt (sodium acetate)



Why? - This shows acetic acid reacting with sodium bicarbonate to produce sodium acetate, carbon dioxide, and water which is balanced. It can be seen from the reaction that the number of atoms of each element is the same on both sides of the equation, adhering to the law of conservation of mass

3. B. The fire gives off heat, light, and smoke

Why? - When wood burns, you can observe light, smoke and the fire give off heat. These are the indicators of a chemical reaction.

4. C. It can be used as a cleaning agent or to unclog drains

Why?- The reaction produces carbon dioxide gas and water, which creates bubbling and fizzing that can help dislodge clogs. This acid-base reaction is commonly used as a natural cleaning agent and to

5. You are part of a community project addressing the harmful effects of traditional combustion-based cooking (e.g., using wood or charcoal). Which of the following options best reflects a scientifically sound and sustainable recommendation for reducing harmful emissions while maintaining efficiency, and why?

- A. Switch to electric stoves powered by coal plants, as they eliminate smoke inside the home.
- B. Use kerosene stoves, because kerosene burns faster and produces less soot than wood.
- C. Adopt solar-powered or biogas stoves, as they minimize incomplete combustion and reduce carbon emissions.
- D. Continue using wood but ensure it's dry, since dry wood burns more easily and with a hotter flame.



Figure 5. cooking using using wood or charcoal (Zulu et al., 2024)

Investigating Chemical Reactions

Complete the table.

Practical Scenario	Reactants	Product	Observable Change	Type of Reaction	Balanced Chemical Equation
Vehicle that run on gasoline					

unclog drains because the gas bubbles help break up blockages

5. C. *Adopt solar-powered or biogas stoves, as they minimize incomplete combustion and reduce carbon emissions.*

Why?- Adoption of solar-powered or biogas stoves minimize incomplete combustion and reduce carbon emissions, thus lowering harmful emissions while maintaining cooking efficiency. Switching to electric stoves powered by coal or using kerosene still involves fossil fuels and emissions. Using dry wood reduces smoke but does not eliminate harmful emissions

Practical Scenario	Reactants	Product	Observable Change	Type of Reaction	Balanced Chemical Equation
Vehicle that runs on gasoline	Hydrocarbons (e.g., C_8H_{18}) + O_2	$CO_2 + H_2O$	Heat and light (combustion), exhaust gases	Combustion	$2C_8H_{18} + 25O_2 \rightarrow 16CO_2 + 18H_2O$
Formation of ammonia in manufacturing fertilizer	$N_2 + H_2$	NH_3	Gas formation (ammonia), pressure drop	Synthesis	$N_2 + 3H_2 \rightleftharpoons 2NH_3$

Formation of ammonia in manufacturing fertilizer						<table><tr><th>Practical Scenario</th><th>Reactants</th><th>Product</th><th>Observable Change</th><th>Type of Reaction</th><th>Balanced Chemical Equation</th></tr><tr><td>Acid rain (HCl) on metal surfaces</td><td>Metal (e.g., Zn or Fe) + Hydrochloric acid (HCl)</td><td>Metal chloride (e.g., ZnCl₂ or FeCl₂) + Hydrogen gas (H₂)</td><td>Metal surface dissolves, bubbles of gas form, metal corrodes or disappears</td><td>Acid-metal reaction / Single displacement</td><td>Zn (s) + 2HCl (aq) → ZnCl₂ (aq) + H₂ (g)</td></tr></table>	Practical Scenario	Reactants	Product	Observable Change	Type of Reaction	Balanced Chemical Equation	Acid rain (HCl) on metal surfaces	Metal (e.g., Zn or Fe) + Hydrochloric acid (HCl)	Metal chloride (e.g., ZnCl ₂ or FeCl ₂) + Hydrogen gas (H ₂)	Metal surface dissolves, bubbles of gas form, metal corrodes or disappears	Acid-metal reaction / Single displacement	Zn (s) + 2HCl (aq) → ZnCl ₂ (aq) + H ₂ (g)
Practical Scenario	Reactants	Product	Observable Change	Type of Reaction	Balanced Chemical Equation													
Acid rain (HCl) on metal surfaces	Metal (e.g., Zn or Fe) + Hydrochloric acid (HCl)	Metal chloride (e.g., ZnCl ₂ or FeCl ₂) + Hydrogen gas (H ₂)	Metal surface dissolves, bubbles of gas form, metal corrodes or disappears	Acid-metal reaction / Single displacement	Zn (s) + 2HCl (aq) → ZnCl ₂ (aq) + H ₂ (g)													
Acid rain (HCl) in metal surfaces (Zn)																		
VI. REFLECTION (Feedback and Continuous Improvement)						This section presents the key highlights and challenges encountered by both teachers and learners during the teaching-learning process throughout the unit. It also includes the adjustments made by the teacher to improve instruction.												

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Appendix A.

Anticipation Guide: Types of Chemical Reactions and Their Real-Life Relevance

Instructions: Before the lesson, read each statement and place a ✓ under the “Before” column if you believe the statement is true, or an ✗ if you believe it is false. After the lesson, revisit the statements and mark them again in the “After” column based on what you have learned. Then discuss the correct answers with your group or class.

		Before (✓/✗)	After (✓/✗)
1	All chemical reactions result in observable changes like color change or production of gas.		
2	Combustion is a type of chemical reaction that occurs when a substance reacts with oxygen.		
3	Rusting of iron is a decomposition reaction.		
4	You can see bubbles occurring when zinc reacts with hydrochloric acid because hydrogen gas is being produced.		
5	Vinegar and baking soda react in a single replacement reaction.		
6	The metal-acid reaction yields a salt and hydrogen gas		
7	Digestive tablets made of magnesium can also react with acid in the stomach, yielding hydrogen gas.		
8	An example of metal-acid reaction is real-life corrosion of metals in acid rain		
9	An example of a chemical reaction is using bleach to clean stains		
10	Understanding about chemical reactions will make you able to make better choices regarding the chemicals in your household products.		

Appendix B.
Rubric for Reaction Station

Criteria	4 – Excellent	3 – Good	2 – Fair	1 – Needs Improvement
Safety and Preparation	Always follows safety rules; fully prepared with all materials and setup.	Follows safety rules with 1–2 reminders; mostly prepared.	Needs several reminders; setup is incomplete or missing items.	Frequently forgets safety rules; and unprepared.
Following Procedures	Performs all steps correctly without help	Performs most steps correctly	Needs help to follow some steps	Has difficulty following steps
Observation	Writes accurate observations	Writes mostly accurate observations.	Writes general or partly wrong observations.	Observations are missing, vague, or wrong.
Explanation	Clearly explain observed changes and relate to a specific type of chemical reaction	Provides a mostly accurate explanation of how the changes relate to a specific type of chemical reaction	Explanation is unclear or partly wrong	Explanation is wrong, missing, or lacks science terms.
Participation and Teamwork	Members are involved and helpful all the time.	Members participates most of the time; works well with others.	Member/s is/are sometimes involved; teamwork is inconsistent.	Members are rarely involved; some may disrupt or avoid group tasks.

Appendix C.
Rubric for POE Experiment

Criteria	4 – Excellent	3 – Good	2 – Fair	1 – Needs Improvement
Safety and Preparation	Always follows safety rules; fully prepared with all materials and setup.	Follows safety rules with 1–2 reminders; mostly prepared.	Needs several reminders; setup is incomplete or missing items.	Frequently forgets safety rules; and unprepared.
Following Procedures	Performs all steps correctly without help	Performs most steps correctly	Needs help to follow some steps	Has difficulty following steps
Prediction	Provides 2 or more scientifically accurate predictions	Gives 1 reasonable prediction with some science basis.	Gives 1 unclear or partially correct prediction.	No prediction or the prediction is incorrect or unrelated.
Observation	Writes accurate observations	Writes mostly accurate observations.	Writes general or partly wrong observations.	Observations are missing, vague, or wrong.
Explanation	Clearly explain observed changes and relate to a specific type of chemical reaction	Provides a mostly accurate explanation of how the changes relate to a specific type of chemical reaction	Explanation is unclear or partly wrong	Explanation is wrong, missing, or lacks science terms.
Participation and Teamwork	Members are involved and helpful all the time.	Members participates most of the time; works well with others.	Member/s is/are sometimes involved; teamwork is inconsistent.	Members are rarely involved; some may disrupt or avoid group tasks.

Appendix D.
Finding Practical Application (Option 1)

Reaction Detective: “Where do you see these reactions in daily life?”
Individual Activity

Chemical reactions are occurring constantly around you, from the cooking of your food, the cleaning of your home, to the rusting of metal. Through this exercise, you will get to investigate a household or personal care product to understand how its chemical ingredients interact. You will also observe some typical household activities that demonstrate various types of chemical reactions in progress, and reflect on what these reactions imply for the environment.

Instructions:

A. Selection:

1. Identify a common household/personal care product, such as: Shampoo, Toothpaste, Bleach, Vinegar, Air freshener, Baking soda, Muriatic acid
2. Identify process/activities at home (where the household/personal care product was used) that demonstrate the type of reactions

B. Task

1. Read about the product to describe the process/activities at home where the product was used.
2. Answer the following:
 - a. What happen when you use this product?
 - b. What type of chemical reaction is it involved in the use of the product? (e.g., neutralization, decomposition)?
 - c. What is the importance of the activity/process that uses of the product to our daily life?

C. Valuing

1. Explain benefits and potential risk of using them (product) or doing them (process/activities)

Output: a simple presentation with visuals/slides that contain the following:

1. Selected product
2. Household Process/Activity (with description)
3. Answers to questions

Rubric:

Criteria	3 – Excellent	2 – Good	1 – Poor
Selection	the activity involving the product was described well	The description of the activity involving the product is limited	Product or activity unclear or missing
Explanation of Reaction	Correctly identifies and explains reaction	Some explanation but missing details	Reaction unclear or not explained
Importance to Daily Life	Clearly explains why it matters	Gives a basic reason	No or unclear explanation
Presentation & Visuals	Neat, complete, and helpful visuals	Mostly clear visuals	hard to read, or missing visuals

Appendix E.
Finding Practical Application (Option 2)

The Art of Chemical Reaction
Individual Activity

Objective:

Students will use acid-base reactions to create visual effects for creative artworks.

Materials:

Baking soda (base)
Vinegar (acid)
Food coloring or liquid watercolor
Small containers or cups
Dropper or pipette
Watercolor paper or thick paper
Spoon or brush

Procedure:

1. Prepare a piece of watercolor paper.
2. Spread a thin layer of baking soda on the paper using a spoon.
3. Mix food coloring or watercolor with vinegar in small cups (1 color per cup).
4. Using droppers or pipettes, drip the colored vinegar onto the baking soda layer. As the vinegar hits the baking soda, it fizzes and bubbles. The reaction spreads the color and creates unique patterns.
5. Let the paper dry. The textures and colors left behind become part of the final art.

Guide question:

1. How did the reaction affect your design?

2. How does understanding chemical reactions enhance creativity?

Source: Kelly. 2020. "Fizzy Art - Baking Soda Paint." Typically Simple. April 20, 2020. <https://typicallysimple.com/fizzy-art-baking-soda-paint/>.

Rubric:

Criteria	3 – Excellent	2 – Good	1 – Poor
Creativity	colors and patterns stand out	nice colors and patterns	design is simple or unclear
Visual Impact of the Art	Final artwork is neat, vibrant, and clearly shaped by the chemical reaction.	Artwork is complete and shows clear reaction effects, though some areas may be inconsistent.	Artwork is partially complete or reaction effects are less visible.
Understanding of Concept	Clearly explains how the reaction works	Gives a basic explanation	Has trouble explaining the reaction
Reflection Answers	Gives thoughtful, detailed answers	Answers clearly with some insight	Answers are simple or unclear

Appendix F. Additional Activity (Take home)

ChemVlog (Individual/Group Activity)

Through this video blog making activity, you will get to investigate some typical household activities that demonstrate various types of chemical reactions at home that demonstrate the type of reactions they are involved in, and their environmental implications.

Instructions:

1. **Selection:** Observe three activities from your home (e.g., cooking, cleaning, or heating). Describe what happens and identify the type of chemical reaction involved.
2. **Video Creation:** Create a 3–5 minute vlog that includes:
 - Activity observed
 - demonstration or reenactment of the reaction (if safe and possible) (*you may include members of the family*)
 - Explain the type of chemical reaction observed
 - Describe the visible signs that help you how you identified the type of reaction.
 - For groupwork, provide a report of the roles and contributions of the members of the group together with the video. Mention the assigned role such as researcher, videographer, presenter, scriptwriter. Show how the group worked together to plan, film and edit the video.

Tips: Creative visuals, props, or analogies to aid understanding

Safety measure:

Handle household chemical like vinegar, baking soda and other carefully. Keep sharp objects like nails, knife and other away from children. Perform reaction that might produce spills over a sink or outside.

Rubric:

Criteria	4	3	2	1
1. Description of Activities	Clearly describes 3 relevant household activities and correctly names the chemical reactions.	Describes 3 good activities with mostly correct reaction names.	Describes some activities but reactions are partly wrong or unclear.	Activities are unclear, not relevant, or reactions are wrong.
2. Scientific Explanation	Clearly explain observed changes and relate to a specific type of chemical reaction	Provides a mostly accurate explanation of how the changes relate to a specific	Explanation is unclear or partly wrong	Explanation is wrong, missing, or lacks science terms.

		type of chemical reaction		
3. Creativity & Presentation	Very creative and interesting vlog with good visuals or props.	Presentation is clear and somewhat creative with some visuals or props.	Simple presentation with little creativity or visuals.	Vlog is not interesting and lacks creativity or visuals.
4. Demonstration & Safety	Demonstrations are safe, clear, and show the reactions well.	Mostly safe demonstrations that show the reactions.	Demonstrations are unclear or may have some safety issues.	Demonstrations are unsafe, unclear, or missing.
5. Teamwork & Contribution (group work only)	Members are involved and helpful all the time.	Members participates most of the time; works well with others.	Member/s is/are sometimes involved; teamwork is inconsistent.	Members are rarely involved; some may disrupt or avoid group tasks.